

Dr. Rakesh Bhaskaran Nair

Surface Engineering, Diagnostics and Materials Intelligence

Engineer against degradation. Detect damage as it develops. Predict integrity before failure.

Principal Investigator and Early-Career Research Leader | Dublin City University

Engineer | Degrade | Detect | Predict | Decide

29	1,260+	h-index 19	18	EUR 220k+
Journal articles	Citations	Google Scholar	First/corr. author	PI/Co-PI funding

RESEARCH PROGRAMME

A - Surface Engineering & Degradation Science

Advanced coatings, repair systems and interfaces for CMAS, erosion, corrosion, wear, oxidation and thermal degradation.

B - Real-Time Diagnostics & NDT

Pyrometry, thermography, acoustic emission, optical sensing and nondestructive evaluation for material-state evolution.

C - Materials Intelligence & Prognostics

Physics-informed analytics, multimodal sensing and machine learning for degradation prediction and integrity decisions.

SELECTED PUBLICATIONS

- Spectral Emissivity Hysteresis in NiTi Powders. Optics & Laser Technology, 2026.
- Nickel-Dependent Emissivity Calibration of NiTi Powders. Measurement, 2025.
- CMAS Corrosion Attack on Thermally Sprayed TBCs. npj Materials Degradation, 2024.
- HEA Coatings Deposited by Thermal Spraying: A Review. Metals, 2023.
- Enhanced Wear Resistance of AlCoCrFeMo Coatings via Thermal Spray. Surface & Coatings Technology, 2024.

ACADEMIC POSITIONING

Materials and manufacturing researcher working at the intersection of surface engineering, extreme-environment degradation, real-time diagnostics and materials intelligence. The programme connects degradation-resistant surfaces, thermal/optical/acoustic sensing and physics-informed analytics.

Flagship PI Project

Principal Investigator - Research Ireland ARC Hub

OptiSIFT: powder degradation, materials-state interpretation and evidence-based reuse or rejection decisions | EUR 185,000 | 2026-2027

PROGRAMME LOGIC

- Engineer and process coatings, repair systems and advanced materials.
- Degrade under CMAS, erosion, corrosion, wear, oxidation, thermal cycling and service-like environments.
- Detect and predict material state to support continued use, repair, reuse, process adjustment or replacement.

TEACHING & SUPERVISION FIT

- MSc teaching: IR Sensors & Thermal Diagnostics for Additive Manufacturing; Thermal Spray Manufacturing Technologies.
- Teachable areas: surface engineering; materials characterisation; AM diagnostics; tribology and failure analysis.
- Mentored 9+ postgraduate researchers and trained 10+ HQPs on laboratory safety and SOPs.

AVAILABLE FOR

Academic collaborations | Invited lectures | FDPs | MSc teaching | Curriculum development | External PhD/MSc co-supervision | Visiting/adjunct opportunities | Fellowship hosting

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